

Tableau de la dérivée



	$f'(n)$	Exemple
a	0	$f(n) = 4 \Rightarrow f'(n) = 0$
$an + b$	a	$f(n) = 3n + 2 \Rightarrow f'(n) = 3$
x^n	$n x^{n-1}$	$f(n) = x^3 \Rightarrow f'(n) = 3x^2$
$\frac{1}{x}$	$-\frac{1}{x^2}$	$f(n) = \frac{1}{x} \Rightarrow f'(n) = -\frac{1}{x^2}$
$\sqrt{x}$	$\frac{1}{2\sqrt{x}}$	$f(n) = 5\sqrt{x} \Rightarrow f'(n) = \frac{5}{2\sqrt{x}}$
$\sqrt[n]{x}$	$\frac{1}{n x^{1/n}}$	$f(n) = \sqrt{x^2 + 2} \Rightarrow f'(n) = \frac{2x}{2\sqrt{x^2 + 2}}$
$\frac{1}{x^n}$	$-\frac{n}{x^{n+1}}$	$f(n) = \frac{1}{x^3} \Rightarrow f'(n) = -\frac{3}{x^4}$
$\cos(n)$	$-\sin(x)$	$f(n) = \cos(x) \Rightarrow f'(n) = -\sin(x)$
$\sin(n)$	$\cos(n)$	$f(n) = \sin(n) \Rightarrow f'(n) = \cos(n)$
$\sqrt[n]{x}$	$\frac{1}{3\sqrt[n]{x^2}}$	$f(n) = \sqrt[3]{x}$ or $\sqrt[3]{x} = x^{\frac{1}{3}}$ $\Rightarrow (x^{\frac{1}{3}})' = \frac{1}{3} x^{\frac{1}{3}-1} = \frac{1}{3} x^{-\frac{2}{3}}$ $\Rightarrow f'(n) = \frac{1}{3 x^{\frac{2}{3}}} = \frac{1}{3\sqrt[3]{x^2}}$



$f'(x)$

Example

$$\frac{1}{(\cos(x))^2}$$

$$\begin{aligned} \tan(x) &= \frac{(\cos(x))^2 + (\sin(x))^2}{(\cos(x))^2} \\ &= \frac{1}{(\cos(x))^2} \end{aligned}$$

$(\sin(u))^r$

$$u^r \cos(u)$$

$$\begin{aligned} f(x) &= \sin(x)^r \\ \Rightarrow f'(x) &= 5x^4 \cos(x^r) \end{aligned}$$

$\cos(u)$

$$-u^r \sin(u)$$

$$\begin{aligned} f(x) &= \cos(x^3) \\ \Rightarrow f'(x) &= -3x^2 \sin(x^3) \end{aligned}$$

a^u

$$+u^r a^u \ln a$$

$$\begin{aligned} f(x) &= 3x^r \\ \Rightarrow f'(x) &= 5x^4 3x^r \ln 3 \end{aligned}$$

$\log_a(u)$

$$\frac{u^r}{u \cdot \ln a}$$

$$\begin{aligned} f(x) &= \log_5(x^2+3) \\ \Rightarrow f'(x) &= \frac{2x}{(x^2+3) \ln 5} \end{aligned}$$

$\arcsin(x)$

$$\frac{1}{\sqrt{1-x^2}}$$

$$f(x) = \arcsin(x) \Rightarrow f'(x) = \frac{1}{\sqrt{1-x^2}}$$

$\arcsin(u)$

$$\frac{u^r}{\sqrt{1-u^2}}$$

$$f(x) = \arcsin(x^2-3) \Rightarrow f'(x) = \frac{2x}{\sqrt{1-(x^2-3)^2}}$$

$\arccos(x)$

$$\frac{-1}{\sqrt{1-x^2}}$$

$$f(x) = \arccos(x) \Rightarrow f'(x) = \frac{-1}{\sqrt{1-x^2}}$$

$\arccos(u)$

$$\frac{-u^r}{\sqrt{1-u^2}}$$

$$f(x) = \arccos(2x) \Rightarrow f'(x) = \frac{-2}{\sqrt{1-4x^2}}$$

$\arctan(x)$

$$\frac{1}{1+x^2}$$

$$f(x) = \arctan(x) \Rightarrow f'(x) = \frac{1}{1+x^2}$$

$\arctan(u)$

$$\frac{u^r}{1+u^2}$$

$$f(x) = \arctan(x^3) \Rightarrow f'(x) = \frac{3x^2}{1+x^6}$$



$f(x)$	$f'(x)$	Example
	$u' + v'$	$f(x) = 2x^2 + 3x$ $\Rightarrow f'(x) = 4x + 3$
$u \times v$	$u'v + v'u$	$(x\sqrt{x^2-n})' = (x)' \sqrt{x^2-n} + (\sqrt{x^2-n})' x$ $= \sqrt{x^2-n} + \frac{2x-1}{2\sqrt{x^2-n}} \cdot x$
f^n	$nf' \cdot f^{n-1}$	$f(x) = (x+1)^2$ $\Rightarrow f'(x) = 2(x+1)' \cdot (x+1)^{2-1}$ $= 2x + 2$
$\frac{u}{v}$	$\frac{u'v - v'u}{(v)^2}$	$(\frac{x^2-1}{x^2})' = \frac{(x^2-1)'x - (x)'(x^2-1)}{x^2}$ $= \frac{2x^2 - x^2 + 1}{x^2} = \frac{x^2 + 1}{x^2}$
$\cos(ax+b)$	$-a \sin(ax+b)$	$(\cos(3x))' = -3 \sin(3x)$
$\sin(ax+b)$	$a \cos(ax+b)$	$(\sin(2x+1))' = 2 \cos(2x+1)$
$e^{u(x)}$	$u'(x) e^{u(x)}$	$(e^{3x})' = 3e^{3x}$
$\ln(u(x))'$	$\frac{u'(x)}{u(x)}$	$\ln(3x+1)' = \frac{3}{3x+1}$
e^n	e^n	$(e^n)' = e^n$
$\ln(x)$	$\frac{1}{x}$	$(\ln x)' = \frac{1}{x}$
$\tan(u)$	$\frac{u'}{(\cos(u))^2}$	$\tan(3x) = \frac{3}{(\cos(3x))^2}$